Three-player modal agents in a public-goods game

Lindheim, Björn

1 Introduction

Setting

Modal agents decide what to do by asking what can be proved about the other players, written with the provability box $\Box$ of the logic GL [1]. Mennen [22] set up a framework for more than two players; here we study three players concretely. The game is a public-goods game: each player contributes ($a_i = 1$) or not, and earns $u_i = 2\sum_j a_j - 3a_i$, so contributing costs 1 net but gives each other player 2. An agent sees p_1, p_2 ("opponent j contributes in this game") and probes q_1, q_2 ("opponent j contributes when I am replaced by a defector"). Behaviour is read off in the long run (tail semantics). An agent is *safe* if it never earns less than either opponent, whoever they are.

Result

- **The three-player FairBot is safe.** $F = \Box(p_1 \wedge p_2)$ (contribute iff it is provable that both others contribute) never earns less than either opponent against any two modal opponents, contributes with two copies of itself and defects against a defector.
- **It is the smallest.** No agent smaller than F (size 4) both cooperates with copies of itself and is safe.
- **All small safe cooperators are FairBot-like.** Exactly 66 agents up to size 6 cooperate with themselves and are safe; none of them uses the probes, and for each GL proves that it contributes only if $\Box^3(p_1 \wedge p_2)$. Every other agent of that size fails against some pair of opponents of size at most 3.
- **A prudent three-player agent needs the probe.** $E = \Box((p_1 \wedge p_2) \wedge (\Box\bot \vee \neg q_1))$ (size 10) cooperates with itself, is safe, and defects against two unconditional contributors; no agent of size at most 7 has all three properties.
- **The definition of the probe matters.** For five basic three-player bots we certify all 125 triples. Changing the probe from "only I am replaced by

a defector" to "everyone except the queried opponent is replaced" changes the outcome of 9 of the 125 triples.

Why it matters

Program equilibrium with modal agents has been worked out almost only for two players. In a three-player public-goods game the natural FairBot generalisation stays safe, is the smallest safe cooperator, and is essentially the only kind up to size 6. Exploiting unconditional contributors requires the probe, as with PrudentBot in two-player games, and the exact definition of the probe is a real modelling choice, since outcomes change with it.

Proved vs checked

The safety of F against all modal opponents is proved in Lean 4 under tail semantics. The minimality, the 66-agent classification, the properties of E and the 125-triple tables are exact computations over all agents up to the stated sizes in the stated language, with GL certificates.

Novelty statement

We believe this result is new. Our prior-art search (2026-10-20) covered:

- Mennen, "Modal combat for games other than the prisoner's dilemma" (LessWrong / Alignment Forum, 2017): framework for modal agents in games with any number of players and actions (player types, ranks, opponent tests naming a target player, an agent to test against and an action); existence and Pareto theorems; no agent families with index laws, no enumeration.
- Gram_Stone, "Modal Chicken" (LessWrong, 2015) and its comments (Alex-Mennen, SilentCal): SwerveBot, NerveBot, CarefulBot, AstuteBot; informal claims about graded CarefulBots and BoldBots.
- Hobson, "Yet More Modal Combat" (LessWrong / Alignment Forum, 2021): a two-system PD bot; generalisation to Chicken and more players left open.
- Critch, Dennis & Russell, "Cooperative and uncooperative institution designs: surprises and problems in open-source game theory" (arXiv 2208.07006): two-player bounded proof-search agents.
- Todorov, ten Napel & Müller, "Parametric Open Source Games" (arXiv 2606.27068): non-modal parametric agents; Chicken and Stag Hunt thresholds.
- Barasz et al. 2014 (arXiv 1401.5577); LaVictoire et al. 2014; Critch, parametric bounded Löb (arXiv 1602.04184); Oesterheld 2022 (arXiv 2211.05057);

LessWrong / Alignment Forum wiki "Modal combat"; web searches for modal or provability-based agents in Chicken, Stag Hunt and multiplayer program equilibrium.

We found no earlier statement of it, but we can't be certain. If you know of prior work, please open an issue.

This result extends Mennen's framework for multiplayer modal agents, whose opponent tests fix one definition of probe atoms (Mennen 2017). Our prior-art search (2026-10-20) covered:

- Mennen, "Modal combat for games other than the prisoner's dilemma" (LessWrong / Alignment Forum, 2017): framework for modal agents in games with any number of players and actions (player types, ranks, opponent tests naming a target player, an agent to test against and an action); existence and Pareto theorems; no agent families with index laws, no enumeration.
- Gram_Stone, "Modal Chicken" (LessWrong, 2015) and its comments (Alex-Mennen, SilentCal): SwerveBot, NerveBot, CarefulBot, AstuteBot; informal claims about graded CarefulBots and BoldBots.
- Hobson, "Yet More Modal Combat" (LessWrong / Alignment Forum, 2021): a two-system PD bot; generalisation to Chicken and more players left open.
- Critch, Dennis & Russell, "Cooperative and uncooperative institution designs: surprises and problems in open-source game theory" (arXiv 2208.07006): two-player bounded proof-search agents.
- Todorov, ten Napel & Müller, "Parametric Open Source Games" (arXiv 2606.27068): non-modal parametric agents; Chicken and Stag Hunt thresholds.
- Barasz et al. 2014 (arXiv 1401.5577); LaVictoire et al. 2014; Critch, parametric bounded Löb (arXiv 1602.04184); Oesterheld 2022 (arXiv 2211.05057); LessWrong / Alignment Forum wiki "Modal combat"; web searches for modal or provability-based agents in Chicken, Stag Hunt and multiplayer program equilibrium.

If you know of related work we missed, please open an issue.

How it was produced

This result was found by AI research agents (the autoresearch harness). It was partly checked by scripts that run Lean proofs (claims K14, K16, K18, K19) and partly checked by script (claims K13, K21); the proof and check files are included so anyone can re-run them.

2 Claims

Claim K13

For five three-player bots $D, C, F = \Box(p_1 \wedge p_2), O = \Box(p_1 \vee p_2), Q = \Box(q_1 \wedge q_2)$, 125 triples are GL-certified under evaluator-replacement probes: $FFF = 111$, $FFD = 000$, $QQQ = 000$; exactly D, F, Q never earn less than either opponent within this five-bot set.

Status: derived ů author check only.

Verified.

- `checks/K13-three_player.py`: 125 triples, all 3000 simultaneous replacement-subset equations GL-certified; 375 flipped-candidate controls; headline cells and finite payoff/self-play classifications. — pass
- `checks/K13-probe_semantics_alternative.py`: Re-certifies the original 125 triples and alternative 125 triples; 6000 equations and 750 flip controls; exactly nine stated differences. — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K14

In the three-player public-goods game $u_i = 2\sum_j a_j - 3a_i$, $F = \Box(p_1 \wedge p_2)$ never earns less than either opponent against any two modal opponents (Lean, tail semantics), while cooperating with two copies and defecting against a defector.

Status: derived ů author check only.

Verified.

- `checks/K14-c046_xreview_k13.py`: (4) Lean 4 core fair3_tail (any opponent traces, X->Y&Z) and fair3_payoff (decided over 8 tails), no sorryAx, false variant rejected; (2) FFF=111, FFD=000, QQQ=000, FCD=010, QBC=101 GL-certified on all 24 replacement equations; (1) 125 K13 triples re-solved independently — pass (Lean-proved, no sorryAx)

Evidence.

- reasoning: If F contributes in the tail, both opponents contributed one height earlier, still in the tail; otherwise F defects and a defector earns at least as much as anyone.

Prior art. Possibly new: no prior art found in the owner's search.

Claim K16

In the guarded three-player language with node-count size and evaluator-replacement probes, size 4 is minimal for self-cooperation plus never earning less than either opponent in $u_i = 2\sum_j a_j - 3a_i$: $F = \Box(p_1 \wedge p_2)$ attains it (tail semantics).

Status: derived ů author check only.

Verified.

- `checks/K16-three_player_minimal.py`: Exhaustive 25 raw guarded trees through size 3; 15625 augmented triples, 375000 GL equations, 46875 flip controls; each smaller self-cooperator has a disadvantage witness; size-4 FairBot self-cooperates and passes finite safety. — pass
- `checks/K16-three_player_minimal.py`: 25 exhaustive raw trees through size 3; 15625 triples / 375000 GL equations / 46875 flip controls; smaller self-cooperators have disadvantage witnesses; embedded K14 Lean fair3_tail and fair3_payoff prove universal upper bound, no sorryAx, false variant rejected. — pass (Lean-proved, no sorryAx)

Evidence.

- reasoning: The exhaustive finite witnesses rule out universal safety for every smaller self-cooperator. K14 proves universal safety of size-4 conjunction FairBot, supplying the matching upper bound.

Prior art. Possibly new: no prior art found in the owner's search.

Claim K18

In the guarded three-player language (evaluator-replacement probes, $u_i = 2\sum_j a_j - 3a_i$), exactly 66 trees of size ≤ 6 self-cooperate and never earn less than either opponent; all are q-free and GL proves $A \to \Box^3(p_1 \wedge p_2)$; others fail vs size-$\leq 3$ pairs.

Status: derived ů author check only.

Verified.

- `checks/K18-c046_safe6.py`: (1) 1,6,18,112,596,3756 guarded trees; (2) self-cooperators 11,41,263,1448; survivors 0,0,0,2,8,56, all others beaten by size<=3 pairs (constants except 8 named trees); (3) survivors q-free, GL proves A->[][][](p1&p2), control rejected; (4) 3438 triples GL-certified, 24 equations each, flips rejected; (5) Lean lag_tail, fair3_payoff, no sorryAx, false variant rejected — pass (Lean-proved, no sorryAx)

Evidence.

- reasoning: The height chain with arbitrary atom valuations is a GL model, so A-\>\[\]^3(p1&p2) gives: x h implies y g and z g for g+3\<=h; lag_tail then yields tail X-\>Y&Z, and a defector never earns less than anyone.

Prior art. Possibly new: no prior art found in the owner's search.

Claim K19

In K18's three-player language, $E = \Box((p_1 \wedge p_2) \wedge (\Box\bot \vee \neg q_1))$ (size 10) self-cooperates, defects against two constant contributors (payoffs 4,1,1) and never earns less than either opponent; no tree of size ≤ 7 has all three properties.

Status: derived ů author check only.

Verified.

- `checks/K19-c046_prudent3.py`: (1) 9 E-cells (EEE=111, ECC=011, ECD=010, EDD=000, E FB FB=111, E E FB=111, E C FB=010, E D FB=000, EEC=111) GL-certified, 24 equations each, flips rejected; (2) GL proves E->[](p1&p2), control rejected; (3) sizes<=7: 1,6,18,112,596,3756,23636 trees, self-coop & defect vs (C,C) 0,0,4,4,68,220,1808, each beaten by a size<=3 pair; (5) Lean lag_tail, fair3_payoff, no sorryAx, false variant rejected — pass (Lean-proved, no sorryAx)
- `checks/K19-c046_xreview_k20.py`: K20 cells incl. RRR=000 (q1->p1) GL-certified on 24 equations, flips rejected; GL proves A<->E with 8 mixed triples 111; GL rejects A<->R (probe essential) — pass

Evidence.

- reasoning: GL soundness on the height chain turns E-\>\[\](p1&p2) into: E contributes at h implies both opponents contribute at all g\<h; lag_tail with k=1 gives tail X-\>Y&Z, and a defector never earns less than anyone.

Prior art. Possibly new: no prior art found in the owner's search.

Claim K21

For K13s five bots, replacing the evaluator alone versus replacing all players other than the queried opponent changes exactly 9 of 125 triples: permutations of (Q, C, O) and (Q, O, O) change from 111 to only Q defecting (GL-certified).

Status: derived ů author check only.

Verified.

- `checks/K13-probe_semantics_alternative.py`: 250 triples, 6000 GL equations, 750 flips; exactly the specified nine differences, QOC 111 versus 011; all probe-free triples unchanged. — pass

Prior art. Partial prior art.

Provenance. Run `20261019-1400`, claims K13, K14, K16, K18, K19, K21, run-folder commit `24c2a64bf8b8cb8be184b4d449ce11068d8270ea`, published 2026-10-20.

3 Discussion

The results use one language (two "contributes" atoms and two probes), one payoff function and tail semantics; other probe definitions change some outcomes, as shown. The classifications stop at sizes 6 to 10. Open questions are the minimum size of a safe self-cooperator that needs the probe (between 8 and 10) and whether these patterns extend to more players or other multiplayer games.

4 How to verify

Each finding has its check scripts in `claims/three-player-modal-agents/checks/` of the findings repository. Run each with `python -I <script>`; it passes if it exits with status 0. The checks are Python scripts and need the packages of the research image (the image's `requirements.txt`); the Lean checks also need `lean`.

5 Revisions

- 2026-10-21 — Production note: Lean proofs run inside check scripts are now described as such, separately from harness-built proofs

References

[1] Mihaly Barasz, Paul Christiano, Benja Fallenstein, Marcello Herreshoff, Patrick LaVictoire, and Eliezer Yudkowsky. Robust cooperation in the prisoner's dilemma: Program equilibrium via provability logic, 2014.

[2] Paul Beame and Søren Riis. More on the relative strength of counting principles. In *Proof Complexity and Feasible Arithmetics*, volume 39 of *DIMACS Series in Discrete Mathematics and Theoretical Computer Science*, pages 13–35. American Mathematical Society, 1998.

[3] Christoph Benzmüller. A (simplified) supreme being necessarily exists, says the computer: Computationally explored variants of gödel's ontological argument. In *Proceedings of KR 2020*, 2020.

[4] Christoph Benzmüller. A comment on modal collapse and ultrafilters in gödel's ontological argument. arXiv:2608.07578, 2026.

[5] Christoph Benzmüller. Gödel's and scott's variants of the ontological argument in lean 4 and tptp thf. arXiv:2609.26806, 2026.

[6] Christoph Benzmüller. Proofs without nominals: Gödel's ontological argument, its shallow embedding, and the open questions of the monatshefte notes. arXiv:2609.36279v4, 2026.

[7] Christoph Benzmüller and David Fuenmayor. Computer-supported analysis of positive properties, ultrafilters and modal collapse in variants of gödel's ontological argument. arXiv:1910.08955, 2019.

[8] Christoph Benzmüller and Lawrence C. Paulson. Quantified multimodal logics in simple type theory. *Logica Universalis*, 7(1), 2013.

[9] Christoph Benzmüller and Dana S. Scott. Notes on gödel's and scott's variants of the ontological argument. *Monatshefte für Mathematik*, 2025.

[10] Christoph Benzmüller, Leon Weber, and Bruno Woltzenlogel Paleo. Computer-assisted analysis of the anderson–hájek ontological controversy. *Logica Universalis*, 2017.

[11] Patrick Blackburn, Maarten de Rijke, and Yde Venema. *Modal Logic*. Cambridge University Press, 2001.

[12] George Boolos. *The Logic of Provability*. Cambridge University Press, 1993.

[13] Andrew Critch. A parametric, resource-bounded generalization of löb's theorem, and a robust cooperation criterion for open-source game theory. *Journal of Symbolic Logic*, 2019.

[14] Andrew Critch, Michael Dennis, and Stuart Russell. Cooperative and uncooperative institution designs: Surprises and problems in open-source game theory. arXiv:2208.07006, 2022.

[15] Kit Fine. Propositional quantifiers in modal logic. *Theoria*, 36(3), 1970.

[16] Melvin Fitting. *Types, Tableaus, and Gödel's God*. Kluwer, 2002.

[17] Murdoch J. Gabbay and Dominic P. Mulligan. Nominal Henkin semantics: simply-typed lambda-calculus models in nominal sets. arXiv:1111.0089, 2011.

[18] Gram_Stone. Modal chicken. LessWrong, 2015.

[19] Donald Hobson. Yet more modal combat. LessWrong / AI Alignment Forum, 2021.

[20] Wesley H. Holliday. A note on algebraic semantics for S5 with propositional quantifiers. *Notre Dame Journal of Formal Logic*, 60(2), 2019.

[21] Patrick LaVictoire, Benja Fallenstein, Eliezer Yudkowsky, Mihaly Barasz, Paul Christiano, and Marcello Herreshoff. Program equilibrium in the prisoner's dilemma via löb's theorem. In *AAAI Workshop on Multiagent Interaction without Prior Coordination*, 2014.

[22] Alex Mennen. Modal combat for games other than the prisoner's dilemma. LessWrong / AI Alignment Forum, 2017.

[23] Mladen Mika and Jakob Nordström. A generalized method for proving polynomial calculus degree lower bounds. In *30th Conference on Computational Complexity (CCC 2015)*, volume 33 of *LIPIcs*, pages 467–487, 2015.

[24] Reinhard Muskens. Intensional models for the theory of types. *Journal of Symbolic Logic*, 72(1), 2007.

[25] Caspar Oesterheld. A note on the compatibility of different robust program equilibria of the prisoner's dilemma, 2022.

[26] Alexander A. Razborov. Proof complexity of pigeonhole principles. In *Developments in Language Theory (DLT 2001)*, volume 2295 of *Lecture Notes in Computer Science*, pages 100–116. Springer, 2002.

[27] Søren Riis. *Independence in Bounded Arithmetic*. PhD thesis, University of Oxford, 1993.

[28] Søren Riis and Meera Sitharam. Non-constant degree lower bounds imply linear degree lower bounds. Technical Report TR97-048, Electronic Colloquium on Computational Complexity, 1997.

[29] Robert M. Solovay. Provability interpretations of modal logic. *Israel Journal of Mathematics*, 25:287–304, 1976.

[30] Aleksandar Todorov, Jesse ten Napel, and Alexander Müller. Parametric open source games. arXiv:2606.27068, 2026.

[31] transhumanist_atom_understander. Is there only one FairBot? LessWrong, with a comment by Zac Hatfield-Dodds, 2026.